

Eigen Stuff

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1 Introduction

Firstly, there is a lot of terminology surrounding this topic, and honestly depending on where your looking they likely won't even be using it. So rather than confusing ourselves with terminology over solution spaces and what not I recommend focusing on the material directly related to Eigenvalues and Eigenvectors. As such these notes will only cover examples of this, but will define the rest of the terminology. I will add an astricts (*) on all the definitions I recommend you pay special attention to.

Okay, so what even are Eigenvalues, Eigenvectors, and Eigen spaces? And how am I going to used them in engineering?

Eigen spaces and solution space are similar and will be explained in definitions. But simply put they are the sets that come out of the consequence that Eigenvalues require directionality out of their corresponding Eigenvector but not magnitude which allows for many many choices of Eigenvector to correspond to any given Eigenvalue

Now in short an Eigenvalue is a scalar multiple in a vector space that when multiplied executes the same transformation on a corresponding Eigenvector as preforming a given linear transformation.

Lets break that down, a linear transform is a mapping (or function) from one vector space to another which obeys certain properties covered in part 3 of the linear algebra introduction material.

Thus an Eigenvalue is the amount and Eigenvector is scaled under a given linear transformation. More an Eigenvector encodes a direction that remains unchanged under the given linear transformation.

2 Definitions and Theorems

Eigenvectors and Eigenvalues

Let V be a vector space and $f : V \rightarrow V$ be a linear mapping. Then λ is called an Eigenvalue to the corresponding Eigenvector v iff

$$f(v) = \lambda v$$

Solution spaces

A solution space of a linear system is the set of all possible solutions to that linear system.

Eigen Spaces

An Eigen Space is the set of all Eigenvectors corresponding to a given Eigenvalue coupled with the zero vector.

Characteristic Polynomial

Let λ be an eigen value of a matrix A . Then the characteristic polynomial is $\det(A - \lambda I) = 0$.

3 Examples

Lets look at the matrix

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$$

To find the Eigenvalues we are going to employ the characteristic polynomial. So

$$\begin{aligned}
0 &= \det\left(\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} - \lambda I\right) \\
&= \det\left(\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \\
&= \det\left(\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) \\
0 &= \det\left(\begin{bmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{bmatrix}\right)
\end{aligned}$$

So we have the polynomial

$$\begin{aligned}
0 &= \det\left(\begin{bmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{bmatrix}\right) \\
&= (1-\lambda)(4-\lambda) - (1)(-2) \\
&= \lambda^2 - 5\lambda + 4 + 2 \\
&= \lambda^2 - 5\lambda + 6 \\
&= (\lambda - 2)(\lambda - 3)
\end{aligned}$$

So we have Eigenvalues 2 and 3.

Lets now calculate some Eigenvectors!

Applying the definition of Eigenvalues and Eigenvectors

$$\begin{bmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

plugging in our values of λ first $\lambda = 2$,

$$\begin{bmatrix} 1-2 & 1 \\ -2 & 4-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Applying the replacement row operation $R_2 \rightarrow R_2 - 2R_1$

$$\begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\cong \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which yields two things. First notice you CAN apply row operations to a non augmented matrix! Just ensure you apply the same operation across both sides of the equation and leave the x,y,z vector alone, just like in an augmented matrix. Second we have the relation $-x + y = 0 \implies x = y$. so we have our first Eigenvector

$$\begin{bmatrix} x \\ x \end{bmatrix}$$

so lets let $x = \pi$ because why not! Then our first Eigenvalue Eigenvector pair is

$$2, \begin{bmatrix} \pi \\ \pi \end{bmatrix}$$

Now I leave it to you to calculate the second but the relation you will get to check is $-2x = y$.

Applying this relation I am going to let $x = e$ but your free to chose whatever you like as all values of x will be in the solution space, so go ahead and be boring and chose 1 if you want. But I am fun and interesting and that's why I chose things like π and e .

That said, zero is a bad choice as that will lead to the trivial solution. This is in our solution set, but woes snot span

it, so we don't have a representation of our solution set.

This gives our full solution

$$2, \begin{bmatrix} \pi \\ \pi \end{bmatrix}$$

and

$$3, \begin{bmatrix} -2e \\ e \end{bmatrix}$$